1. (a)

maximize Δ

subject to
$$X_{23} + X_{24} - X_{12} = 0$$
$$X_{13} + X_{23} - X_{34} = 0$$
$$-X_{12} - X_{13} + \Delta = 0$$
$$X_{24} + X_{34} - \Delta = 0$$
$$X_{ij} \le C_{ij} \quad \forall i,j \in E$$
$$X_{ij} \ge 0 \quad \forall i,j \in E$$

where $C = (C_{ij})_{4\times 4} = \begin{bmatrix} 0 & 8 & 7 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 12 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$E = \{(1,2),(1,3),(2,3),(2,4),(3,4)\}$

```python
import cvxpy as cp

x12 = cp.Variable()
x13 = cp.Variable()
x23 = cp.Variable()
x24 = cp.Variable()
x34 = cp.Variable()
delta = cp.Variable()

constraints = [
    x23 + x24 - x12 == 0,
    x13 + x23 - x34 == 0,
    delta - x12 - x13 == 0,
    x24 + x34 - delta == 0,
    x12 >= 0, x12 <=8,
    x13 >= 0, x13 <= 7,
    x23 >= 0, x23 <= 2,
    x24 >= 0, x24 <= 4,
    x34 >= 0, x34 <= 12
]

obj = delta

problem = cp.Problem(cp.Maximize(obj),constraints)
problem.solve()

print(f"x12 = {x12.value:.2f}")
print(f"x13 = {x13.value:.2f}")
print(f"x23 = {x23.value:.2f}")
print(f"x24 = {x24.value:.2f}")
print(f"x34 = {x34.value:.2f}")
print(f"optimal_value = {obj.value:.2f}")
```

```
x12 = 6.00
x13 = 7.00
x23 = 2.00
x24 = 4.00
x34 = 9.00
optimal_value = 13.00
```

(b) Dual Problem:

minimize $\sum_{1\le i,j\le 4} C_{ij} Z_{ij}$

subject to
$$Z_{ij} \ge Y_i - Y_j, \quad \forall i,j \in E$$
$$Y_1 - Y_4 = 1$$
$$Z_{ij} \ge 0, \quad \forall i,j \in E$$

Complementarity conditions:
$$X_{ij}\left[Z_{ij} - (Y_i - Y_j)\right] = 0, \quad \forall i,j \in E$$
$$Z_{ij}(C_{ij} - X_{ij}) = 0, \quad \forall i,j \in E$$

That is:

$X_{12}(Z_{12} - 1 + Y_2) = 0$ $\quad$ $Z_{12}(8 - X_{12}) = 0$

$X_{13}(Z_{13} - 1 + Y_3) = 0$ $\quad$ $Z_{13}(7 - X_{13}) = 0$

$X_{23}(Z_{23} - Y_2 + Y_3) = 0$ $\quad$ $Z_{23}(2 - X_{23}) = 0$

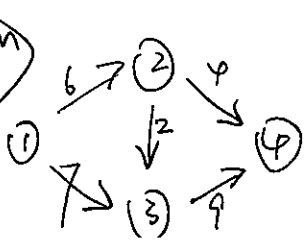
$X_{24}(Z_{24} - Y_2) = 0$ $\quad$ $Z_{24}(4 - X_{24}) = 0$

$X_{34}(Z_{34} - Y_3) = 0$ $\quad$ $Z_{34}(12 - X_{34}) = 0$

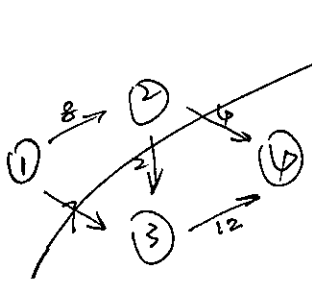
optimal value: 13

$\therefore [X_{12}, X_{13}, X_{23}, X_{24}, X_{34}] = [6, 7, 2, 4, 9]$, $[Z_{12}, Z_{13}, Z_{23}, Z_{24}, Z_{34}] = [0, 1, 1, 1, 0]$

(c) maximum flow



minimum cut



2.

(a) The optimal solution: $X^* = [5 \ 15 \ 0 \ 0 \ 0]^T$

The optimal value of the original problem: 175

(b) $A_B^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{10}{3} \\ 0 & 0 & 1 \end{bmatrix}$

$$X_B^* + \lambda A_B^{-1} e_1 = \begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} + \lambda A_B^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 + \frac{\lambda}{3} \\ 10 - \frac{2}{3}\lambda \\ 5 \end{bmatrix} \geq 0$$

$\therefore \lambda \in [-45, 15]$, b_1 can be changed to: $[5, 65]$

(c) $60 \in [5, 65]$ $\therefore$ The optimal basis is still $\{2, 4, 1\}$

$X_B = A_B^{-1} \tilde{b} = \begin{bmatrix} \frac{55}{3} \\ \frac{10}{3} \\ 5 \end{bmatrix}$ $\therefore X^* = [5, \frac{55}{3}, 0, \frac{10}{3}, 0]^T$

New optimal value: $\frac{625}{3}$

(d) $2 \in B$

$$C_N^T - \lambda e_2^T A_B^{-1} A_N = [0, 0] - [\lambda - 10, 0, -5] \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{10}{3} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{10}{3} - \frac{\lambda}{3} \\ \frac{5}{3} + \frac{\lambda}{3} \end{bmatrix} \geq 0$$

$\therefore \lambda \in [-5, 10]$, C_2 can be changed to: $[5, 20]$